

Exact Standard-Channel Rank of a Signed-Reversal Incidence Layer

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Abstract

We study a noncentral signed-reversal incidence layer in the signed permutation group. For the diagonal layer $\tilde{B}I_n(k, k)$, the aggregate operators on the two ambient standard channels have exact ranks $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$ whenever the layer is nonempty. The rank numbers are classical matching-scheme multiplicities; the object-specific content is the collapse of the signed-reversal layer onto the reversal idempotent and the positivity of the resulting scalars. We further show that the same two ranks persist on the type-D parity sublayer $\tilde{D}I_n(k, k) = \tilde{B}I_n(k, k) \cap D_n$: a sign-character split reduces the restriction to a sign-weighted correction scalar, which is controlled by an explicit factorized generating function, and, in the odd pair-standard case, by a direct type-D trace identity. We also record a positive-only separator add-on: for $X_n = \tilde{B}I_n(1, 1)$ and $n \geq 4$, the layer is neither a signed-poset linear-extension set nor an exact left-translate tiler of B_n .

Keywords: signed permutations; hyperoctahedral group; reversal incidence layers; type-D parity sublayer; association schemes; rook polynomials; signed posets.

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1 Introduction

This paper studies a signed analogue of the type-A one-incidence layer. In type A, the layer is defined by one hit on the identity diagonal and one hit on the reflected diagonal determined by the reversal permutation. Here the ambient group is the signed permutation group B_n , and the signed layer records positive hits on the identity diagonal and on the reversal diagonal. The reversal diagonal is essential: it is the geometric second diagonal that makes the object noncentral but still rigid enough to be controlled by the matching scheme of the reversal pairs.

The main theorem is deliberately scoped. We do not attempt a full native B_n Fourier fingerprint or a general Weyl-group theorem. We prove the exact ranks of two ambient standard-channel aggregate operators on every nonempty diagonal row $\tilde{B}I_n(k, k)$. The numbers that appear, $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$, are classical multiplicities from the matching scheme attached to the reversal involution. The object-specific contribution is that this noncentral signed-reversal layer collapses onto the corresponding matching-scheme components, and that the remaining scalars are strictly positive. We then extend the two rank equalities to the type-D parity sublayer $\tilde{D}I_n(k, k) = \tilde{B}I_n(k, k) \cap D_n$, where $D_n = \ker \chi$ is the even-sign subgroup: a sign-character split writes the type-D aggregate as the average of the sign-blind and the sign-weighted aggregates, and the sign-weighted correction is controlled by a factorized generating function and, for the odd pair-standard channel, by a direct type-D trace identity.

There are two independent additions around the main theorem. First, the positive-only row $X_n = \tilde{B}I_n(1, 1)$ is separated from two natural chamber families: for $n \geq 4$ it is neither a Reiner-style signed-poset linear-extension set nor an exact left-translate tiler of B_n . Second, the appendix records finite $n = 8$ native B_n fingerprint evidence. That evidence explains the known zero rows in one finite atlas slice, but it is not promoted to a classification theorem.

The proof of the rank theorem has three parts. The even rows collapse directly by centralizer symmetry and a permanent-domination positivity lemma. The odd rows reduce to two residual scalar sequences, called H_m and K_m . These residuals are identified with signed-rook moment formulae and then proved positive by exact order-three recurrences and ratio induction. The computational material in the repository verifies the polynomial identities and finite edge rows using exact integer or rational arithmetic; the paper states the mathematical identities that those certificates check.

Structure. Section 2 fixes the signed-reversal layer and states the main theorem. Section 3 proves the positive-only separator theorem. Sections 4–9 prove the rank theorem. Section 10 extends it to the type-D parity sublayer. Section 11 records the claim boundary. The appendices give the exact replay routing and the finite native B_n fingerprint boundary.

2 Definitions and Main Statement

Let B_n be the signed permutation group, written as pairs (π, ε) with $\pi \in S_n$ and $\varepsilon_i \in \{\pm 1\}$. We use zero-indexed coordinates and set

$$\rho(i) = n - 1 - i.$$

Define

$$\tilde{B}I_n(k, \ell) = \{(\pi, \varepsilon) \in B_n : \#\{i : \pi(i) = i, \varepsilon_i = +1\} = k, \#\{i : \pi(i) = \rho(i), \varepsilon_i = +1\} = \ell\}.$$

The theorem concerns only the diagonal layer $\tilde{B}I_n(k, k)$ and only the two ambient standard-channel aggregate operators M_{ref} and $M_{\text{pair-std}}$.

Theorem 1 (Main theorem). *For every $n \geq 2$ and every $k \geq 1$ such that $\tilde{B}I_n(k, k)$ is nonempty,*

$$\text{rank } M_{\text{ref}}(\tilde{B}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil, \quad \text{rank } M_{\text{pair-std}}(\tilde{B}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil - 1.$$

The same two rank equalities hold on the type-D parity sublayer $\tilde{D}I_n(k, k) = \tilde{B}I_n(k, k) \cap D_n$; this is Theorem 3 in Section 10.

Remark 1 (Boundary). *No full B_n fingerprint theorem, enumeration novelty claim, or general Weyl theorem is asserted. The positive-fixed marginal is treated as classical signed-permutation enumeration; in particular the positive-fixed zero row is the type-B derangement sequence [OEI26, CZ07]. The rank number and the reversal projector are matching-scheme bookkeeping [Del73, BI84, GR01]. The contribution here is the collapse and scalar positivity for the specific noncentral layer $\tilde{B}I_n(k, k)$ and its type-D parity sublayer, plus the separate separator theorem below for $\tilde{B}I_n(1, 1)$.*

3 Positive-Only Separator Add-On

Set

$$X_n = \tilde{B}I_n(1, 1).$$

This section uses the equivalent one-indexed signed-word convention $w = (w_1, \dots, w_n)$, with $|w_1|, \dots, |w_n|$ a permutation of $\{1, \dots, n\}$ and $\rho(i) = n + 1 - i$. Thus

$$X_n = \{w \in B_n : \#\{i : w_i = i\} = 1, \#\{i : w_i = \rho(i)\} = 1\}.$$

For a signed word define the Reiner-style total order [Rei93]

$$T(w) : -w_n < \dots < -w_1 < 0 < w_1 < \dots < w_n.$$

Let $\Sigma_n = \{-n, \dots, -1, 0, 1, \dots, n\}$. For $X \subseteq B_n$, put

$$C(X) = \bigcap_{w \in X} \{(a, b) \in \Sigma_n^2 : a \text{ precedes } b \text{ in } T(w)\}.$$

If $X = L_B(P)$ is the signed linear-extension set of a signed poset P , then every strict relation of P , and every relation in its transitive closure, lies in $C(X)$.

Theorem 2 (Positive-only separator). *For every $n \geq 4$, the layer $X_n = \tilde{B}I_n(1, 1)$ is not a Reiner-style signed-poset linear-extension set and is not an exact left-translate tiler of B_n . For $n = 3$, the divisibility obstruction disappears and there is an exact left-translate tiling of B_3 by X_3 .*

Part	Proof type	Replay route
signed-poset obstruction	local sign-extension construction plus exact finite $n = 4$ bridge	S12A
$n = 3$ tiling exception	exact finite cover	S12B
translate-tiler obstruction	rook-polynomial count, divisibility, tail inequalities, and finite window	S12B

Proof-route map. This is an additional scoped theorem, separate from the two-standard-channel rank theorem.

Proof. We first prove the signed-poset obstruction. For $n \geq 5$, any prescribed signs on at most two absolute labels can be extended to an element of X_n . Indeed, positive labels create fixed or reversal hits only when placed in their two dangerous rows, while negative labels create no positive hit. Place the prescribed positive labels, then add exactly one positive fixed hit and one positive reversal hit; all remaining labels are placed negatively. If the odd center is used, it is used as the intended center hit, since the center supplies both hit types at once. Otherwise auxiliary positive labels are chosen away from the center and away from their dangerous rows. With at most two prescribed labels, there is always such a spare choice for $n \geq 5$. The odd center is the only place where $\rho(i) = i$ can interfere with the sign-extension construction; the construction either uses it deliberately as the single center hit or avoids it when choosing the auxiliary positive labels.

Given any ordered pair $a \neq b$ in Σ_n , choose signs so that b lies before 0 and a lies after 0 in $T(w)$. If a, b are nonzero and $a \neq -b$, prescribe $|a|$ with the sign of a and $|b|$ with the opposite sign to b . If $a = -b$, prescribe only $|a|$ with the sign of a . If one entry is 0, prescribe the other label on the required side of 0. The sign-extension construction then gives $w \in X_n$ in which b precedes a . Hence $C(X_n) = \emptyset$ for $n \geq 5$.

The boundary case $n = 4$ is finite: an exact finite check records witnesses in X_4 reversing all 72 ordered pairs in Σ_4 . Thus $C(X_4) = \emptyset$ as well. If $X_n = L_B(P)$ for some signed poset P , all relations of P would lie in $C(X_n)$, so P would be the antichain and $L_B(P) = B_n$. But $X_n \neq B_n$, for instance the all-negative identity has no positive fixed or reversal hit. This proves the signed-poset obstruction for all $n \geq 4$.

For the tiler obstruction, an exact left-translate tiling by X_n requires $N_n = |X_n|$ to divide $G_n = |B_n| = 2^n n!$. The count of X_n is obtained from the rook polynomial

$$R_{2m}(A, B, Z) = (1 + 2AZ + 2BZ + A^2Z^2 + B^2Z^2)^m,$$

$$R_{2m+1}(A, B, Z) = R_{2m}(A, B, Z)(1 + AZ + BZ + ABZ),$$

where A marks selected positive fixed events, B selected positive reversal events, and Z distinct forced cells. Inclusion-exclusion for exactly one event of each type gives

$$N_n = \sum_{p, q \geq 1, u} (-1)^{p+q} p q [A^p B^q Z^u] R_n(A, B, Z) 2^{n-u} (n-u)!.$$

Equivalently, for

$$S_n(Z) = (A\partial_A)(B\partial_B)R_n(A, B, Z)|_{A=B=-1},$$

we have

$$\frac{N_n}{G_n} = \sum_u \frac{[Z^u]S_n(Z)}{2^u(n)_u}.$$

The differentiated series are

$$S_{2m}(Z) = 4m(m-1)Z^2(1-Z)^2(1-4Z+2Z^2)^{m-2},$$

$$S_{2m+1}(Z) = Z(1-4Z+2Z^2)^m + 4m(m-1)Z^2(1-Z)^3(1-4Z+2Z^2)^{m-2}.$$

Substituting $Z = -x$ makes the tails alternating. The certificate `P15_S12_GATE_B_TILER_CERTIFICATE`, routed in Appendix A, records the exact finite window $4 \leq n \leq 21$, the $n = 3$ exact cover, and the symbolic all- n tail inequalities below.

For even $n = 2m \geq 22$, write

$$S_{2m}(-x) = 4m(m-1)x^2(1+x)^2f(x)^{m-2}, \quad f(x) = 1 + 4x + 2x^2.$$

Because $f(x) = (1 + (2 + \sqrt{2})x)(1 + (2 - \sqrt{2})x)$ has positive linear factors, the coefficients of $(1+x)^2f(x)^{m-2}$ are ultra-log-concave by Newton's inequalities. If $c_u = [x^u]S_{2m}(-x)$, this gives

$$\frac{c_{u+1}}{2^{u+1}(2m)_{u+1}} \leq \frac{c_u}{2^u(2m)_u},$$

so the normalized alternating tail is decreasing. The certificate then checks the two finite symbolic inequalities $L_9 > 1/11$ and $U_6 < 1/10$ after the shift $m = t + 11$; all coefficients of the resulting polynomials are positive.

For odd $n = 2m + 1 \geq 23$, the first coefficient of $S_{2m+1}(-x)$ is negative and all later coefficients are positive. Indeed

$$S_{2m+1}(-x) = xf(x)^{m-2}Q_m(x),$$

and the certificate lists the coefficients of Q_m : the nonconstant coefficients are positive for $m \geq 11$, while the top three support degrees are supplied by the positive high coefficients of Q_m . To prove the normalized tail is monotone from degree 2, put $C_m(x) = S_{2m+1}(-x)$ and use the equivalent coefficient positivity of

$$H_m(x) = 2((2m+1)C_m(x) - xC'_m(x)) - C_m(x)/x.$$

The identity $H_m(x) = f(x)^{m-3}P_m(x)$ is checked symbolically, and Newton log-concavity for $f(x)^{m-3}$ reduces the possible negative contribution of the linear term of P_m to a positive quadratic lower bound for every $m \geq 11$. As in the even case, the certificate checks $L_9 > 1/11$ and $U_6 < 1/10$ after the shift $m = t + 11$ by coefficient positivity. Thus the alternating-tail bounds and exact finite checks give

$$9 < G_4/N_4 < 10,$$

$$10 < G_n/N_n < 11 \quad (n \geq 5 \text{ odd}),$$

$$11 < G_n/N_n < 12 \quad (6 \leq n \leq 20 \text{ even}),$$

$$10 < G_n/N_n < 11 \quad (n \geq 22 \text{ even}).$$

Thus G_n/N_n is never an integer for $n \geq 4$, so X_n cannot tile B_n by exact left translates. For $n = 3$, an exact finite check records an exact cover by 24 left translates of the two-element set X_3 . $\square$

4 Proof Strategy and Matching-Scheme Background

Let P_ρ be the permutation matrix of the reversal involution and put

$$P_+ = \frac{I + P_\rho}{2}.$$

The centralizer of ρ in S_n acts on ordered coordinate pairs. For even $n = 2m$, its orbital algebra on coordinates is the matching, or cocktail-party, association scheme with basis $\{I, P_\rho, J - I - P_\rho\}$. For odd $n = 2m + 1$, the center coordinate adds the usual center row and column orbitals, but the non-center pair block is the same matching-scheme block. We use only the standard consequences that P_+ projects onto the +1 eigenspace of P_ρ , that

$$\text{rank } P_+ = \left\lceil \frac{n}{2} \right\rceil,$$

and that the corresponding pair-standard quotient has dimension $\lceil n/2 \rceil - 1$. These are classical association-scheme facts; the proof below supplies the layer-specific collapse onto these spaces and proves that the relevant scalars are nonzero.

Lemma 1 (Even collapse and positivity). *For even $n = 2m$ and every nonempty diagonal row $\tilde{B}I_{2m}(k, k)$ with $k \geq 1$, the reference-channel aggregate collapses to a positive scalar multiple of P_+ , and the pair-standard aggregate collapses to a positive scalar on the corresponding matching-scheme standard quotient. Hence the two even-row ranks are m and $m - 1$.*

Proof. The row is invariant under the centralizer of ρ , so the coordinate aggregates lie in the matching-scheme orbital algebra. In the reference channel, every generic off-pair orbit has a sign-reversing involution, while the diagonal and reversal orbits are equal by reversal symmetry. Thus only the P_+ component remains. The same block cancellation on ordered pair coordinates leaves only the pair-standard quotient component. The surviving scalar is a signed permanent coefficient with one distinguished matching contribution. Write the relevant board matrix as a nonnegative distinguished part plus the remaining signed-orbit contribution. In the expansion of the permanent, the distinguished matching term is present whenever the diagonal row is nonempty; every other term is paired by the sign-reversing involutions used above or contributes nonnegatively after the orbit collapse. Thus the distinguished term dominates the collapsed scalar and gives strict positivity. The same domination applies on the pair-standard quotient after the block cancellation, so the ranks are the classical multiplicities of the two matching-scheme components. $\square$

Lemma 2 (Odd structural reduction). *Let $n = 2m + 1$. The odd reference channel splits as*

$$M_{\text{ref}} = 2aP_+^{\text{pairs}} + a_c E_{\text{center}}.$$

The center split and trace identities reduce all nonempty odd reference rows except $k = 1$ to the even positive scalars and the center scalar. In the odd pair-standard channel, the bridge $\mu = a_c + 4a_{\text{even}}$ and the trace identity reduce all nonempty rows with $k \geq 2$ to the same positive inputs. Thus the only odd residuals are the reference scalar $a(2m + 1, 1) - a(2m, 1)$ and the pair-standard scalar $\lambda'(2m + 1, 1)$.

Proof. Separate the fixed center coordinate from the m non-center reversal pairs. The center coordinate contributes the one-dimensional block E_{center} , while the non-center coordinates remain in the matching-scheme block spanned by the pair idempotent P_+^{pairs} . Taking traces on the diagonal row expresses the non-center reference scalar in terms of the even scalar and the center contribution, except in the $k = 1$ row where the center-nonfixed correction is not eliminated. On the pair-standard side, the non-center pair quotient has two scalar pieces; the displayed bridge identifies one of them with already positive reference/even data, and the trace identity solves for the other for $k \geq 2$. The row $k = 1$ leaves precisely λ' . $\square$

5 Nonempty Diagonal Rows

Lemma 3 (Nonempty positive diagonal condition). *For the theorem range $k \geq 1$, the diagonal layer $\tilde{B}I_n(k, k)$ is nonempty exactly as follows. If $n = 2m$, then*

$$1 \leq k < m, \quad \text{or} \quad k = m \text{ and } m \text{ is even.}$$

If $n = 2m + 1$, then

$$1 \leq k \leq m, \quad \text{or} \quad k = m + 1 \text{ and } m \text{ is even.}$$

The row $k = 0$ is nonempty for every $n \geq 2$, but it is outside the theorem statement.

Proof. Prescribe two positive-hit sets: F for fixed hits and R for reversal hits. Off the odd center, a valid partial prescription must assign at most one image to each domain and must have distinct prescribed images. Thus a non-center position cannot simultaneously be a positive fixed hit and a positive reversal hit.

For $n = 2m$, this gives the upper bound $k \leq m$, since the k fixed-hit domains and k reversal-hit domains are disjoint. For $k < m$, construct examples by reversal-pair blocks. If $k = 2a$, choose a reversal pairs to be fully fixed-positive and a disjoint reversal pairs to be fully reversal-positive. If $k = 2a + 1$, do the same for $2a$ hits and use two further reversal pairs: one supplies a single positive fixed hit and the other supplies a single positive reversal hit. This uses k pairs when k is even and $k + 1$ pairs when k is odd, hence it is possible for every $k < m$.

At the top row $k = m$, all non-center positions are prescribed. If I is the set of positions sent as fixed, the complement is sent as reversed. Injectivity forces $I = \rho(I)$. Hence I is a union of reversal pairs and has even cardinality, so the top row exists exactly when m is even.

For $n = 2m + 1$, the center is fixed and reversed simultaneously. Give the center positive sign to contribute one fixed hit and one reversal hit. If $k = 1$, use the empty non-center prescription; if $k > 1$, apply the even $2m$ construction to the non-center positions with $k - 1$ hits of each kind. This gives exactly $1 \leq k \leq m$, plus the top row $k = m + 1$ when m is even. $\square$

6 Odd Reference Channel: the H_m Residual

For a polynomial $\sum_r h_r t^r$, write

$$\Phi_N \left(\sum_r h_r t^r \right) = \sum_r h_r 2^{N-r} (N-r)!.$$

This is the signed-rook completion functional, a signed version of the classical rook-polynomial completion count [KR46]: after r nonattacking special cells are selected in an N by N remaining board, the unmatched rows and columns can be completed by $2^{N-r} (N-r)!$ signed bijections.

Lemma 4 (H_m is the odd V_{ref} , $k = 1$, residual). *For $m \geq 2$,*

$$a(2m + 1, 1) = a(2m, 1) + H_m.$$

The edge value is $H_2 = 22$. For $m \geq 3$, put

$$A(t) = 1 - 4t + 2t^2$$

and

$$P_m(t) = (2m - 2)(2t - 1)(2mt^3 - (4m + 2)t^2 + (2m + 2)t - 1) - A(t)^2.$$

Then

$$H_m = \Phi_{2m-1} \left((1 - t)A(t)^{m-3} P_m(t) \right).$$

Proof. The odd reference operator splits the center coordinate from the non-center reversal pairs:

$$M_{\text{ref}} = 2aP_+^{\text{pairs}} + a_c E_{\text{center}}.$$

In the $k = 1$ row, the trace identity isolates the non-center scalar as

$$(2m)a(2m+1, 1) = (2m)a(2m, 1) + C_m,$$

where C_m is the center-nonfixed correction. It remains to compute C_m .

Force a non-center source and insert the odd center into one functional edge. The center split gives one fixed target, one reversal target, and $2m - 2$ ordinary targets. After the forced edge is removed, the remaining board has $2m - 1$ rows and columns, so its signed-rook expansion is evaluated by Φ_{2m-1} . The free non-center reversal pairs contribute the factor $A(t)$, and the two local insertion alternatives give

$$H_m = (2m - 2)L_m - B_m,$$

with

$$B_m = \Phi_{2m-1} \left((1 - t)A(t)^{m-1} \right),$$

$$L_m = \Phi_{2m-1} \left((1 - t)(2t - 1)A(t)^{m-3}(2mt^3 - (4m + 2)t^2 + (2m + 2)t - 1) \right).$$

Combining these two expressions gives the stated polynomial $P_m(t)$ and $C_m = 2mH_m$.

The only nonlocal algebra in the derivation is the equality between the expanded edge-insertion count and the compact Φ_{2m-1} expression. This is not a finite extrapolation step. The certificate `P15_S11A_H_TRUE_SCALAR_EDGE_INSERTION_VERIFIER`, routed in Appendix A, checks the local signed-rook edge factors, the transfer from the expanded expression to the displayed univariate target, and the following all- m annihilation.

After the small edge rows $m = 2, 3, 4$ are checked directly, the symbolic residual has the form

$$E_z(\text{local} - (2m - 2)R_m) = A(s)^{m-5}Q_m(s), \quad m \geq 5,$$

where $E_z(F) = F(1, s) - \partial_z F(1, s)$. Under the moment substitution $s = 1/(2x)$, applying integration by parts to

$$I_j = \int_0^\infty e^{-x}(2x^2 - 4x + 1)^{m-5}x^j dx$$

gives the recurrence

$$jI_{j-1} + (-4j - 4m + 15)I_j + (2j + 4m - 12)I_{j+1} - 2I_{j+2} = -\delta_{j0}.$$

The verifier rewrites the transformed residual with this recurrence until only the basis $I_0, I_1, 1$ remains, and all three coefficients are identically zero in $\mathbb{Z}[m]$. Hence the displayed formula is the true scalar residual for every $m \geq 2$. $\square$

7 Odd Pair-Standard Channel: the K_m Residual

Lemma 5 (K_m is the odd pair-standard, $k = 1$, residual). *For odd $n = 2m + 1$, set*

$$K_m = (m - 1)\lambda'(2m + 1, 1).$$

The edge values are $K_2 = 28$ and $K_3 = 14640$. For $m \geq 4$, define

$$B(t) = 2t^2 - 4t + 1$$

and

$$\begin{aligned} L_m(t) = & 2(2m-3)t + 2(-4m^2-8m+19)t^2 + 2(24m^2+18m-68)t^3 \\ & + 2(-56m^2-24m+122)t^4 + 2(56m^2+24m-92)t^5 \\ & + 2(-12m^2-44m+12)t^6 + 2(-16m^2+56m)t^7 + 2(8m^2-24m+8)t^8. \end{aligned}$$

Then

$$\lambda'(2m+1, 1) = \Phi_{2m} \left(B(t)^{m-4} L_m(t) \right),$$

and therefore

$$K_m = (m-1) \Phi_{2m} \left(B(t)^{m-4} L_m(t) \right).$$

Proof. The odd pair-standard channel has two scalar pieces, μ and λ' . The identity $\mu = a_c + 4a_{\text{even}}$ and the trace relation close every nonempty odd row with $k \geq 2$. Thus only the $k = 1$ scalar λ' remains.

Choose a distinguished non-center reversal pair and the odd center. Let j be a generic source outside that pair and outside the center. On the non-center pair-standard line,

$$\lambda' = M_{\text{pair}}[i, i] + M_{\text{pair}}[\rho(i), i] - 2M_{\text{pair}}[j, i].$$

Put $X = u - 1$, $Y = v - 1$, and $Z = uv - 1$, and define

$$\begin{aligned} a &= 1 + 2t(X + Y) + t^2(X^2 + Y^2), & c &= 1 + tZ, \\ sx &= 1 + tX, & sy &= 1 + tY, & h &= 1 + t(X + Y), \\ \text{same} &= (1 + u)sx + (1 + v)sy, & \text{generic} &= 4h^2. \end{aligned}$$

For the two same-pair entries, the forced edge lies inside the distinguished reversal pair. The center contributes c , the unused mate contributes sx or sy , and the other $m - 1$ non-center reversal pairs contribute a ; the two forced signs have weights $1 + u$ and $1 + v$. This gives $ca^{m-1}\text{same}$.

For the generic entry, the forced ordinary edge touches two non-center reversal pairs. Those two pairs leave two hook factors, giving h^2 ; the forced ordinary sign contributes 2, and the coefficient -2 in λ' gives the subtracted factor $\text{generic} = 4h^2$. The center again contributes c , and the remaining $m - 2$ reversal pairs contribute a^{m-2} . Since these row and column sets are disjoint, the rook polynomial is the product of the block factors. Therefore, for every $m \geq 2$,

$$\lambda'(2m+1, 1) = [uv] \Phi_{2m} \left(c \left(a^{m-1}\text{same} - a^{m-2}\text{generic} \right) \right).$$

A forced entry leaves $2m$ rows and columns, so the completion functional is Φ_{2m} .

The case $m = 1$ has no pair-standard quotient. For $m = 2, 3$, the product formula has nonnegative exponents but the univariate $B(t)^{m-4}L_m(t)$ presentation is not used; these are the direct edge rows $K_2 = 28$ and $K_3 = 14640$. For $m \geq 4$, coefficient extraction over $\mathbb{Z}[m, t]$ reduces the bivariate product above to $B(t)^{m-4}L_m(t)$. The public certificate route is split into three exact checks listed in Appendix A: P15_S11A_K_TRUE_SCALAR_BIVARIATE_VERIFIER checks the local true-entry factors, P15_S11A_K_GLOBAL_PRODUCT_PHI_CERTIFICATE checks the product/ Φ accounting over the independent reversal pairs, and P15_S11A_SCALAR_LINK_SYMBOLIC_VERIFIER checks that the bivariate target reduces to the displayed univariate $L_m(t)$. All residual polynomial coefficient lists are zero, so this proves that the displayed K_m sequence is the true pair-standard scalar sequence. $\square$

8 Ratio-Induction Positivity

Lemma 6 (Order-three positivity). *Let U_m denote either H_m or K_m , with an exact order-three recurrence*

$$p_0(m)U_m + p_1(m)U_{m+1} + p_2(m)U_{m+2} + p_3(m)U_{m+3} = 0.$$

Set $\rho_m = U_{m+1}/U_m$. If the required base rows, adjacent base-ratio checks, and polynomial sign checks hold, then $\rho_m \geq 4m^2$ propagates and $U_m > 0$ for all rows in the theorem range.

Proof. The recurrence gives

$$\rho_{m+2} = \frac{-p_2(m) - p_1(m)/\rho_{m+1} - p_0(m)/(\rho_m \rho_{m+1})}{p_3(m)}.$$

For the H_m recurrence, the base rows $m = 2, 3, 4$ have $H_m > 0$ and $\rho_m \geq 4m^2$, and the polynomial signs are $p_0 > 0$, $p_1 > 0$, $-p_2 > 0$, $p_3 > 0$, and

$$D_H(m) = 16m^2(m+1)^2(-p_2) - 4m^2p_1 - p_0 - 64m^2(m+1)^2(m+2)^2p_3 > 0.$$

This is exactly the cleared-denominator inequality for $\rho_{m+2} \geq 4(m+2)^2$ under $\rho_m \geq 4m^2$ and $\rho_{m+1} \geq 4(m+1)^2$.

For the K_m recurrence, the base rows $m = 2, \dots, 8$ have $K_m > 0$ and $\rho_m \geq 4m^2$, and for $m \geq 7$ the polynomial signs are $p_0 > 0$, $p_1 \leq 0$, $-p_2 > 0$, $p_3 > 0$, and

$$D_K(m) = 16m^2(m+1)^2(-p_2) - p_0 - 64m^2(m+1)^2(m+2)^2p_3 > 0.$$

The term $p_1 \leq 0$ can be discarded because it improves the lower bound. The adjacent base ratios ρ_7 and ρ_8 seed the recurrence from $m = 7$. Hence the same induction proves $K_m > 0$ for every $m \geq 2$. The recurrence identities are holonomic-style exact identities in the standard sense of creative telescoping and P-recursive verification [Zei90, PWZ96]; no finite extrapolation is used. In the replay table, P15_S8_HM_RECURRENCE_CERTIFICATE supplies the H_m recurrence, base rows, and sign polynomials, while P15_S9C_KM_RECURRENCE_CERTIFICATE supplies the corresponding K_m data. The inequalities above are exactly the cleared-denominator checks recorded by those certificates. $\square$

9 Proof Assembly

Proof of Theorem 1. The nonempty rows are exactly those of Lemma 3. For even n , Lemma 1 gives the required collapse, positivity, and ranks.

For odd $n = 2m + 1$, Lemma 2 reduces all nonempty rows with $k \geq 2$ to the even positive inputs and center trace identities. In the reference channel, the remaining $k = 1$ scalar is $a(2m, 1) + H_m$. Lemma 4 identifies H_m with the true center-nonfixed residual, and Lemma 6 gives $H_m > 0$.

In the pair-standard channel, the residual $k = 1$ eigenvalue is λ' , equivalently $K_m = (m-1)\lambda'$. Lemma 5 identifies this formula sequence with the true pair-standard scalar, and Lemma 6 gives $K_m > 0$. Thus both ambient standard-channel operators are nonzero scalar multiples of their classical matching-scheme idempotent components. Their ranks are therefore $\lceil n/2 \rceil$ and $\lceil n/2 \rceil - 1$. $\square$

10 The Type-D Parity Sublayer

Let $\chi: B_n \rightarrow \{\pm 1\}$ be the sign character $\chi(\pi, \varepsilon) = \prod_i \varepsilon_i$, and let $D_n = \ker \chi$ be the even-sign subgroup (the type-D reflection group inside B_n). The *type-D parity sublayer* is the intersection

$$\tilde{D}I_n(k, \ell) = \tilde{B}I_n(k, \ell) \cap D_n = \{(\pi, \varepsilon) \in \tilde{B}I_n(k, \ell) : \prod_i \varepsilon_i = +1\}.$$

This section proves that the two ambient standard channels have the *same* exact ranks on the diagonal type-D layer as on the full layer.

Theorem 3 (Type-D rank theorem). *For every $n \geq 2$ and every $k \geq 1$ such that $\tilde{D}I_n(k, k)$ is nonempty,*

$$\text{rank } M_{\text{ref}}(\tilde{D}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil, \quad \text{rank } M_{\text{pair-std}}(\tilde{D}I_n(k, k)) = \left\lceil \frac{n}{2} \right\rceil - 1.$$

As in the full-layer case, each rank equals a fixed matching-scheme multiplicity minus the vanishing of one aggregate scalar; the theorem is the statement that those scalars stay nonzero (indeed strictly positive) after the parity restriction. Write s^D (respectively λ'^D) for the reference (respectively pair-standard) non-center scalar of the type-D layer, and s^B, λ'^B for the corresponding full-layer scalars of Sections 7–9.

10.1 The character split

Lemma 7 (Character split). *For either channel, let $M^B = \sum_{w \in \tilde{B}I_n(k, k)} P_w$ be the full (sign-blind) aggregate and $M^\chi = \sum_{w \in \tilde{B}I_n(k, k)} \chi(w) P_w$ the sign-weighted aggregate, where P_w is the channel representation of w . Then*

$$M(\tilde{D}I_n(k, k)) = \frac{1}{2}(M^B + M^\chi), \quad \text{so} \quad s^D = \frac{1}{2}(s^B + s^\chi), \quad \lambda'^D = \frac{1}{2}(\lambda'^B + \lambda'^\chi).$$

Proof. The indicator of D_n inside B_n is $\frac{1}{2}(1 + \chi)$, since χ is a nontrivial character. Summing $\frac{1}{2}(1 + \chi(w))P_w$ over $w \in \tilde{B}I_n(k, k)$ gives $\frac{1}{2}(M^B + M^\chi)$; passing to the non-center scalar on each channel is linear. $\square$

The full-layer scalars s^B, λ'^B are strictly positive by the main theorem (Sections 7–9). It therefore suffices to control the sign-weighted correction scalars s^χ, λ'^χ .

10.2 The sign-weighted generating function

Mark a signed permutation by $A^{\text{Fix}+}C^{\text{Rev}+}$, recording positive identity hits and positive reversal hits. The reversal partitions the n coordinates into $m = \lfloor n/2 \rfloor$ two-element pairs $\{i, \rho(i)\}$ and, for odd n , one fixed center. Writing $Q = (1 - A)^2 + (1 - C)^2$, the sign-weighted marked count factorizes over the pair decomposition:

$$\sum_{w \in B_n} \chi(w) A^{\text{Fix}+(w)} C^{\text{Rev}+(w)} = \begin{cases} Q^m, & n = 2m, \\ (AC - 1) Q^m, & n = 2m + 1. \end{cases} \quad (1)$$

Indeed, the sign character is multiplicative over the pair blocks; within one pair the four sign patterns cancel against the two off-diagonal placements to leave exactly the block factor Q , and for odd n the center coordinate contributes the scalar $AC - 1$. Specializing the forced-edge construction of Section 7 to the sign-weighted weights turns each channel scalar into a single diagonal coefficient of (1): for $n = 2m + 1$,

$$s^\chi(2m + 1, k) = [A^k C^k] (A^2 - 1)(AC - 1) Q^{m-1}, \quad \lambda'^\chi(2m + 1, k) = [A^k C^k] (AC - 1) Q^m. \quad (2)$$

Setting $S'(a, c) = [A^a C^c] ((1 + A)^2 + (1 + C)^2)^m$, the pair-standard scalar has the closed diagonal form

$$\lambda'^\chi(2m + 1, k) = S'(k - 1, k - 1) - S'(k, k), \quad \lambda'^\chi(2m + 1, 1) = -2^m(m^2 - m - 1). \quad (3)$$

Both closed forms are verified against the exact permanent through $n = 15$ (Appendix A).

10.3 Even n

For $n = 2m$ the sign-weighted generating function (1) is Q^m with Q a sum of two squares, so its diagonal channel scalars are sign-controlled; combined with the even full-layer positivity (Section 4) the split of Lemma 7 gives $s^D, \lambda'^D > 0$, hence the two even ranks. The pair-standard channel is in fact unconditional, exactly as in the full-layer even case.

10.4 Odd n : reference channel

By Lemma 7, $s^D = \frac{1}{2}(s^B + s^X)$ with $s^B > 0$. It remains to show $s^B > |s^X|$ wherever $s^X < 0$.

Lemma 8 (Sign localization). *Write $T(k) = [A^k C^k](1 - A^2)Q_+^{m-1}$ with $Q_+ = (1 + A)^2 + (1 + C)^2$, so that $s^X(2m + 1, k) = T(k) - T(k - 1)$ up to the fixed positive factor of (2). For $3k > 2m$ one has $s^X(2m + 1, k) \geq 0$, with the only exceptions the two cells $(m, k) \in \{(2, 2), (4, 3)\}$.*

Proof sketch. On the band $3k > 2m$ every nonzero summand of $T(k)$ is sign-definite (disjoint support of the two binomial factors), so $T(k) \leq 0$ and the claim reduces to $|T(k)| \leq |T(k - 1)|$. Termwise the ratio of matched summands is bounded by $B = 2(m - k + 1)^2/k^2 < 1$; the only summand of $T(k)$ with no partner in $T(k - 1)$ is a single *orphan* at index $i_0 = m + 1 - k$, present only for the two boundary families $3k \in \{2m + 1, 2m + 2\}$. On those two families the surplus of one matched index $i_0 + 1$ already dominates the orphan: the ratio $(1 - B)|\text{term}(i_0 + 1, k - 1)|/|\text{term}(i_0, k)|$ has the closed rational forms

$$R_A(r) = \frac{(2r^2 - 1)(2r - 1)(2r + 3)}{24(4r - 1)} \quad (m = 3r + 1, k = 2r + 1),$$

$$R_B(r) = \frac{2r^2(r - 1)(2r + 1)}{3(4r - 3)} \quad (m = 3r - 1, k = 2r),$$

and $R_A \geq 1$ for $r \geq 3$, $R_B \geq 1$ for $r \geq 2$, each equivalent to an explicit quartic that is nonnegative by its base value and a positive derivative; the finitely many small cells (including $(2, 2), (4, 3)$) are checked directly. The exact contraction, the closed forms, and the finite census are in Appendix A. $\square$

For the cells with $s^X < 0$ (necessarily $k \leq \frac{2m}{3}$ by Lemma 8, apart from the two exceptions, at which $s^D > 0$ is checked directly) the strict domination $s^B > |s^X|$ follows from a pure type-B counting inequality: with $d_k = |\tilde{B}I_{2m}(k, k)|$,

$$k d_k > 2m |s^X|,$$

which is closed by an explicit reversal-pair gadget embedding $d_k \geq (\text{multinomial}) \cdot D_{m-k}$ together with the Jensen bound $D_r = |\tilde{B}I_{2r}(0, 0)| \geq 4^r(2r)!/4$ and the sign-localization band $k \leq \frac{2m}{3}$ (Appendix A). Hence $s^D > 0$ on every nonempty odd reference row, giving the odd reference rank $\lceil n/2 \rceil$.

10.5 Odd n : pair-standard channel

Here a *direct* type-D trace identity closes the channel without any domination. Fix a diagonal cell (k, k) , $n = 2m + 1$, and let $X^D = \tilde{D}I_{2m+1}(k, k)$. The pair-standard aggregate is the nonnegative integer matrix $M_{\text{pair}}^D[j, i] = \#\{w \in X^D : w(i) = j\}$.

Lemma 9 (Type-D pair-standard trace identity). *With $\mu^D = (n g_c^D - |X^D|)/(n - 1)$ the center scalar ($g_c^D = M_{\text{pair}}^D[c, c]$) and $\text{NF}^D = \sum_{w \in X^D} \#\{i : \pi(i) = i, \varepsilon_i = -1\}$ the total negative-fixed count,*

$$\text{tr } M_{\text{pair}}^D = k|X^D| + \text{NF}^D = |X^D| + (m - 1)\lambda'^D + \mu^D, \quad (4)$$

$$\mu^D \leq |X^D|. \quad (5)$$

Consequently $(m - 1)\lambda'^D = (k - 1)|X^D| + \text{NF}^D - \mu^D \geq (k - 2)|X^D| + \text{NF}^D$.

Proof. The first equality in (4) counts fixed points: each $w \in X^D$ has exactly k positive identity-fixed points, so $\#\text{Fix}(w) = k + \#\{\text{negative-fixed}\}$. The second equality is the orbital block decomposition of the pair-standard operator. Let $H \leq B_n$ be the *sign-free* centralizer of the reversal: the group of unsigned coordinate permutations that commute with ρ and fix the center, so H permutes the m reversal pairs and may swap the two coordinates inside a pair, $H \cong B_m$. Every element of H carries no sign changes, hence $H \subseteq D_n$; the full signed centralizer of ρ also contains the center sign-flip, which is *not* in D_n and is deliberately excluded. Conjugation by H preserves $\tilde{D}I_n(k, k)$ (it fixes the positive identity-hit count, the positive reversal-hit count, and the sign parity) and commutes with M_{pair}^D . The induced action of H on the m reversal-pair indices is the full symmetric group S_m , whose permutation representation splits as trivial $\oplus$ standard; consequently the pair-standard operator has the single non-center eigenvalue λ'^D of multiplicity $m - 1$ (the standard part), together with the center eigenvalue μ^D and the trivial eigenvalue $|X^D|$. This is the same block form as the full-layer operator of Section 7. For (5): every row sum and every column sum of M_{pair}^D equals $|X^D|$ (each w is a permutation), so M_{pair}^D is $|X^D|$ times a doubly stochastic matrix and its real eigenvalue μ^D is at most $|X^D|$. The block form and the trace identity are verified by exact enumeration for $n = 5, 7$ (Appendix A), the same finite standard as Section 7. $\square$

Proposition 1 (Odd pair-standard positivity). $\lambda'^D > 0$ for every nonempty odd diagonal cell.

Proof. For $k \geq 3$, Lemma 9 gives $(m - 1)\lambda'^D \geq (k - 2)|X^D| > 0$. For $k = 2$ the same bound gives $(m - 1)\lambda'^D \geq \text{NF}^D$, which is positive because the explicit element that swaps $1 \leftrightarrow \rho(1)$, fixes every other coordinate, and takes $\varepsilon = +1$ exactly on $\{0, c, 1\}$ lies in $\tilde{D}I_{2m+1}(2, 2)$ (two positive identity hits $\{0, c\}$, two positive reversal hits $\{1, c\}$, an even number $2m - 2$ of negative signs) and has $2m - 3 > 0$ negative-fixed points. For $k = 1$, $\lambda'^X(2m + 1, 1) = -2^m(m^2 - m - 1)$ by (3); setting $K_m = (m - 1)\lambda'^B(2m + 1, 1)$ (Lemma 5, whose displayed sequence is the true scalar by the $S11A-K$ link) and $D_m = (m - 1) \cdot 2^m(m^2 - m - 1)$, the certified ratio $K_{m+1}/K_m \geq 4m^2$ (Section 7, Appendix A) together with the elementary $D_{m+1}/D_m = 2m(m^2 + m - 1)/((m - 1)(m^2 - m - 1)) < 4m^2$ for $m \geq 3$ and the base values $K_2 = 28 > D_2 = 4$, $K_3 = 14640 > D_3 = 80$ give $K_m > D_m$ for all $m \geq 2$, i.e. $\lambda'^B(2m + 1, 1) > |\lambda'^X(2m + 1, 1)|$. Hence $\lambda'^D = \frac{1}{2}(\lambda'^B + \lambda'^X) > 0$. For $n = 3$ (so $m = 1$) the multiplicity $m - 1$ is zero and λ'^D does not occur; the unique nonempty diagonal row is $\tilde{D}I_3(1, 1)$, and its two channel ranks are $(\lceil n/2 \rceil, \lceil n/2 \rceil - 1) = (2, 1)$ by direct computation (Appendix A), exactly as in the full-layer case. $\square$

10.6 Conclusion

Even n gives both type-D ranks directly; for odd n the reference rank follows from $s^D > 0$ and the pair-standard rank from Proposition 1. This proves Theorem 3. The type-D layer thus carries the same two standard-channel ranks as the full signed-reversal layer, with the parity restriction absorbed entirely by the sign-weighted correction scalars s^X, λ'^X of (2).

11 Boundaries and Outlook

The theorem proved here is the scoped two-standard-channel rank theorem for the diagonal signed-reversal layer and its type-D parity sublayer (Section 10). The separator theorem is separately scoped to $X_n = \tilde{B}I_n(1, 1)$. The enumeration of positive fixed points and the matching-scheme multiplicities are used as classical background and are not claimed as new.

No full native B_n fingerprint theorem, all-irrep saturation theorem, or general Weyl-group theorem is asserted. The type-D result is confined to the same two ambient standard channels and the same diagonal cells as the full-layer theorem; no other reflection subgroup or channel is treated. The finite $n = 8$ fingerprint material in Appendix B is included because it clarifies the

boundary of plausible stronger statements: it records exact finite zero mechanisms and selected finite defect accounting, while leaving the remaining atlas rows open.

The natural next mathematical stage is not to broaden the main theorem by wording, but to decide which finite fingerprint mechanisms admit clean all- n statements. Until such mechanisms are proved uniformly, they remain appendix evidence rather than theorem statements.

A Computational Certificates and Reproducibility

All paths in this appendix are relative to the companion repository root. The scripts are deterministic exact-arithmetic checks: they use integer or rational arithmetic and have no floating-point acceptance criterion. Artifact-producing scripts are rerun as

```
python -B <script> -write-json <json>.
```

When a script exposes Markdown output, the replay command also supplies `-write-md <md>`. Report-only commands omit output flags when the script does not expose them. The finite rows in these reports are edge checks or diagnostics; the all- m claims are carried by polynomial identities, integration-by-parts certificates, product/ Φ block accounting, and ratio induction.

How to read the certificates. The certificates have three roles. First, the paper gives the structural reductions and states the scalar identities in mathematical form. Second, the certificate scripts check exact algebraic identities over integer or rational polynomial rings, and exact integration-by-parts or recurrence reductions; these are not finite extrapolations from a long initial segment. Third, finite replay scripts provide edge checks, convention checks, and small-row smoke tests. Thus a line marked “PASS” below should be read as a specific exact identity or finite replay whose mathematical input and output are listed in the adjacent columns.

Claim block	Mathematical content checked	Primary replay artifacts
Even rows	matching-scheme collapse and positive scalar for both standard channels	P15_S4_EVEN_N_THEOREM_CERTIFICATE
Odd reduction	reduction to the two odd $k = 1$ residual scalar sequences	P15_S5_ODD_STRUCTURE_CERTIFICATE
Odd reference residual	H_m true-scalar link, holonomic recurrence, and ratio positivity	P15_S8_HM_RECURRENCE_CERTIFICATE; P15_S11A_H_TRUE_SCALAR_EDGE_INSERTION_VERIFIER
Odd pair-standard residual	K_m true-scalar link, product/ Φ accounting, recurrence, and positivity	P15_S9C_KM_RECURRENCE_CERTIFICATE; P15_S11A_K_TRUE_SCALAR_BIVARIATE_VERIFIER; P15_S11A_K_GLOBAL_PRODUCT_PHI_CERTIFICATE
End-to-end smoke replay	independent exact small-row rank checks against theorem values	P15_S11_RAW_RANK_SMOKE_CHECK_CERTIFICATE
Positive-only separator	signed-poset obstruction and translate-tiler divisibility obstruction	P15_S12_GATE_A_SIGNED_POSET_CERTIFICATE; P15_S12_GATE_B_TILER_CERTIFICATE
Finite native B_n appendix	finite $n = 8$ zero mechanisms and selected extra-defect audits	P15_FULL_BN_*_CERTIFICATE; external $n = 8$ atlas CSVs

Replay	Command script	Artifact basename	Status/role
S3	p15_s3_exact_engine.py	P15_S3_ENGINE_CERTIFICATE	PASS; engine
S4	p15_s4_even_theorem_certificate.py	P15_S4_EVEN_N_THEOREM_CERTIFICATE	PASS; even
S5	p15_s5_odd_structure_certificate.py	P15_S5_ODD_STRUCTURE_CERTIFICATE	PASS; odd split
S6	p15_s6_square_certificate.py	P15_S6_SQUARE_CERTIFICATE	PASS; counts only
S7	p15_s7_independent_red_team.py	P15_S7_INDEPENDENT_RED_TEAM_CERTIFICATE	PASS; replay
S8	p15_s8_hm_recurrence_certificate.py	P15_S8_HM_RECURRENCE_CERTIFICATE	PASS; H_m
S9	p15_s9_odd_pair_reduction_certificate.py	P15_S9_ODD_PAIR_REDUCTION_CERTIFICATE	PASS; pair red.
S9C	p15_s9c_km_recurrence_certificate.py	P15_S9C_KM_RECURRENCE_CERTIFICATE	PASS; K_m

Replay	Command script	Artifact basename	Status/role
S10	p15_s10_public_boundary_review.py	P15_S10_PUBLIC_BOUNDARY_CERTIFICATE	PASS; boundary
S11D	p15_s11_raw_rank_smoke_check.py	P15_S11_RAW_RANK_SMOKE_CHECK_CERTIFICATE	PASS; raw rank
S11A-H	p15_s11a_h_true_scalar_edge_insertion_verifier.py	P15_S11A_H_TRUE_SCALAR_EDGE_INSERTION_VERIFIER	PASS; H link
S11A-transfer	p15_s11a_scalar_link_symbolic_verifier.py	P15_S11A_SCALAR_LINK_SYMBOLIC_VERIFIER	PASS; transfer
S11A-K local	p15_s11a_k_true_scalar_bivariate_verifier.py	P15_S11A_K_TRUE_SCALAR_BIVARIATE_VERIFIER	PASS; local K
S11A-K global	p15_s11a_k_global_product_phi_certificate.py	P15_S11A_K_GLOBAL_PRODUCT_PHI_CERTIFICATE	PASS; global K
S12A	p15_s12_gate_a_signed_poset_certificate.py	P15_S12_GATE_A_SIGNED_POSET_CERTIFICATE	PASS; poset sep.
S12B	p15_s12_gate_b_tiler_certificate.py	P15_S12_GATE_B_TILER_CERTIFICATE	PASS; tiler obs.

The type-D parity sublayer of Section 10 adds the following exact-arithmetic certificates (the replay scripts and their JSON artifacts reside in the `type_d` subdirectory of the companion repository; each script accepts `-no-write/-out` and reports a top-level status field).

Replay	Command script	Artifact basename	Status/role
D-sign	opus_g15_C_closure_verify.py	P15D_G15_C_CLOSURE	PASS; $s^\times$ localization
D-tail	opus_g20_oatail_symbolic_verify.py	P15D_G20_OATAIL_SYMBOLIC	PASS; orphan closed forms
D-pair	opus_g22_psdome_typeD_trace_verify.py	P15D_G22_PSDOM_TYPED_TRACE	PASS; type-D trace
D-form	opus_g16_pairstd_closure_verify.py	P15D_G16_PAIRSTD_CLOSURE	PASS; $\lambda^\times$ closed form

For readability, the internal replay IDs have the following public roles: S11A-H is the odd reference true-scalar link; S11A-K local/global and S11A-transfer are the odd pair-standard true-scalar link; S8 and S9C are the holonomic recurrence/positivity certificates for the two residual sequences; S12A is the signed-poset separator certificate; and S12B is the translate-tiler divisibility certificate. The filenames are stable repository identifiers, not additional mathematical claims.

The rank theorem uses only the scoped two-standard-channel conclusion. The S12 rows add only the separately stated positive-only separator theorem for $X_n = \tilde{B}I_n(1, 1)$, $n \geq 4$. The four type-D rows certify the finite and polynomial checks behind Theorem 3 (the sign-weighted closed forms, the $s^\times$ localization and its orphan-family closed forms, and the type-D trace identity), all on the same two standard channels and the same diagonal cells. The certificate table does not add a full B_n fingerprint theorem, a classification theorem, a general Weyl theorem, or an enumeration novelty claim.

B Finite Native B_n Fingerprint Boundary

The main theorem of this paper is not a full native B_n Fourier fingerprint theorem. We nevertheless record one finite boundary computation because it is useful for interpretation: the naive expectation that every native irreducible block saturates the $R = +$ bound already fails at $n = 8$.

All statements in this appendix section are finite statements for the displayed $n = 8$ atlas. They do not imply an all- n classification, an all-irrep saturation theorem, or a general Johnson-scheme vanishing theorem. The external $n = 8, k \geq 2$ defect census contains 41 fixed-plus family-instances and 114 defect rows. The currently certified accounting covers 6 family-instances and 44 rows, leaving 35 family-instances and 70 rows open.

Slice	fam.	rows	acct. fam.	acct. rows	open fam.	open rows
$n = 8, k = 2$	5	20	1	15	4	5
$n = 8, k = 3$	36	94	5	29	31	65
total	41	114	6	44	35	70

The accounted family-instances are

$$(n, k, \lambda^+) \in \{(8, 2, (1)), (8, 3, (1, 1)), (8, 3, (2)), \\ (8, 3, (4)), (8, 3, (4, 1)), (8, 3, (2, 2, 2, 1, 1))\}.$$

The phrase “accounted” here means finite exact certificate or finite quotient-layer accounting, not an all- n theorem.

Proposition 2 (Pre-Specht cancellation for the $(1, 1)$ zero family). *For $X = \tilde{B}I_8(3, 3)$ and every partition $\nu \vdash 6$,*

$$M_{(1,1)|\nu}(X) = 0.$$

Certified proof sketch. Before applying a complement Specht module, the fixed-plus block with $\lambda^+ = (1, 1)$ is a matrix over the group algebra $\mathbb{Q}[S_6]$. The exact certificate groups the 192 active signed terms into 96 group-algebra keys. For every key there are two surviving completions with the same complement permutation and opposite S_2 sign; their coefficients are -4 and 4 . Thus all 96 keys vanish already in $\mathbb{Q}[S_6]$. Applying any representation S^ν , $\nu \vdash 6$, preserves zero. $\square$

Proposition 3 (Pure-plus Johnson degree-drop). *For $X = \tilde{B}I_8(3, 3)$,*

$$M_{(2,2,2,1,1)|\emptyset}(X) = 0.$$

Certified proof sketch. The conjugate partition of $(2, 2, 2, 1, 1)$ is $(5, 3)$, so

$$S^{(2,2,2,1,1)} \cong S^{(5,3)} \otimes \text{sgn}.$$

After the corresponding sign twist, the certified operator A_{sgn} acts on the 3-subset permutation module $M^{(3)}$. Let U_2 be the pair-incidence submodule spanned by the vectors v_P of 3-subsets containing a fixed pair P . The Johnson certificate verifies the exact containment

$$A_{\text{sgn}}(M^{(3)}) \subseteq U_2.$$

Since the top quotient $M^{(3)}/U_2$ is isomorphic to $S^{(5,3)}$, the sign-twisted operator kills this quotient. Untwisting by the sign representation gives the displayed pure-plus zero. An independent direct rational seminormal Specht computation also gives zero entries throughout the 28-dimensional block. $\square$

Remark 2 (Appendix boundary). *The finite zero mechanisms above explain all twelve zero rows under the $n = 8, k = 3$ atlas census: eleven rows from Proposition 2 and one row from Proposition 3. Other certified finite packages account for selected nonzero extra-defect families, including $T_1^{(8,2)}$, $T_2^{(8,3)}$, $T_4^{(8,3)}$, and $\lambda^+ = (4, 1)$ at $n = 8, k = 3$. The remaining extra-defect rows are not classified here.*

Data and Code Availability

The companion repository contains the LaTeX source, deterministic replay scripts, certified JSON/Markdown artifacts, finite atlas inputs used by the appendix, and a SHA-256 manifest for the public package. The scripts replay the finite checks and exact polynomial identities referenced in Appendix A; the theorem statements are the statements proved in this manuscript.

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